

2. (a) Crude oil is formed from simple marine organisms.

Tick (✓) the box next to the length of time it takes for the organisms to turn into crude oil.

[1]

hundreds of years

☐

thousands of years

☐

millions of years

☐

- (b) Crude oil can be separated into simpler mixtures called fractions.

Tick (✓) the box next to the method used to separate crude oil into fractions.

[1]

fractional distillation

☐

filtration

☐

cracking

☐

polymerisation

☐

(c) The table shows properties of some fractions that are obtained from crude oil.

Fraction	Size of molecules (chain length)	Viscosity	Ease of ignition	Amount of smoke formed
petrol	$C_5 - C_{10}$	very runny	very easy	no smoke
naphtha	$C_8 - C_{12}$	fairly runny	quite easy	little smoke
kerosene	$C_{10} - C_{16}$	thick	quite hard	quite a lot of smoke
diesel oil	$C_{14} - C_{20}$	very thick	very hard	very smoky

Use only the information from the table to answer the following questions.

(i) Name the fraction which is the easiest to pour. [1]

(ii) Name the fraction which is the hardest to burn. [1]

(iii) Name the fraction with the smallest **range** of chain lengths. [1]

.....

(iv) Name the fraction which burns with the cleanest flame. Give the reason for your choice. [2]

Fraction

Reason

